

Précis + autocatpath

Grounded, untiring research collaboration — status

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“Grounded claims, not fluent guesses — every assertion traced to a paragraph, a calculation, or a measurement, and an untiring machine to keep checking.”

Hallucinations: where they come from

- LLMs answer from the training distribution, not from evidence — fluent guesswork whenever the evidence is thin
- Confabulation is measurable: semantic entropy separates “knows” from “making it up”
- Retrieval alone does not fix it — retrieved $\neq$ verified support for the claim actually made

How fast it degrades: the model eats its own output

- Models fed their own generations degrade within a few generations — rare knowledge (the tails) vanishes first
- Collapse is progressive and self-reinforcing without fresh, grounded data
- The antidote is small: keep *accumulating* real data alongside the synthetic — even a modest share — and collapse is averted
- Agentic pipelines recreate the same loop *at inference time*: summaries of summaries, cited as if they were sources

Design consequence: every claim needs an anchor *outside* the model.

State of the art: more hypotheses than anyone can check

- Co-Scientist (Nature '26): multi-agent hypothesis generation, validated in vitro (AML drug repurposing)
- Mechanism: test-time compute + a tournament of *LLM judges* — external grounding stays at the wet lab
- Output is a ranked firehose of hypotheses — **the human becomes the bottleneck**

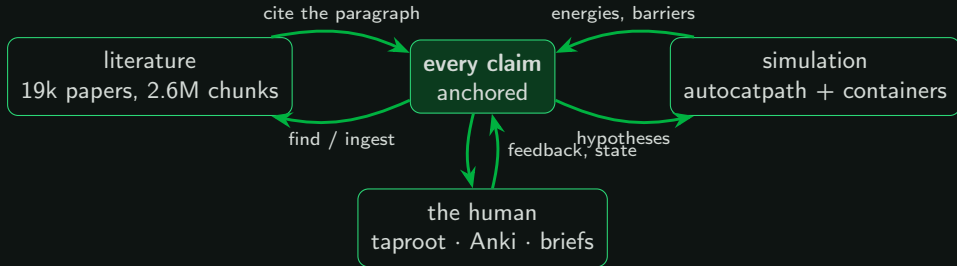
the validation ladder — we fill the middle rungs

literature consistency (précis)

cheap simulation (autocatpath)

wet-lab experiment \$\$\$

Our answer: ground the machine, budget the human



Both collaborators are **bounded-context agents**: the LLM has a token budget, the human has working memory and one screenful of sharp vision. Same discipline for both.

Inside précis: what the system holds

- One verb surface — `get` / `search` / `put` / `edit` / `tag` / `link` — over **54** kinds of thing (39 in active use). Everything below is those six verbs
- **Read**: papers (28,309 tracked, 19,651 ingested), patents, news, web pages, email, encyclopedia entries, transcripts — **2.6M** chunks, each summarized, keyworded, embedded, clustered
- **Written**: drafts, $\text{\LaTeX}$ and markdown, figures, diagrams, slide decks, flashcards (7,264)
- **Modelled**: structures, proteins, materials, reaction routes and pathways — and the hardware kinds: CAD, PCB, parts, datasheets
- **Given** to the agent: **143** skills — “how do I ...”, pulled at the moment of need — plus the kind catalog and the model catalog
- **Grown** by the agent: **9,663** memories (9,222 of them dreamt), **49,403** typed links, 3,453 agent logs — the connections it found, not the ones we gave it
- **Self-ops**: todos (2,202 filed, 120 open), quests, jobs, alerts, gripes — the system files its own bugs

Grounding in literature

- Corpus: **19,651** papers fully ingested (28,309 tracked), **2.6M** searchable chunks, 101 patents
- One substrate: papers, patents, drafts, web pages, notes — everything lands as *chunks*, searchable by meaning, by string, by category, by link (next slide)
- Taproot: claim → supporting papers → *the exact paragraph* (search-to-section)
- Citation-graph crawl keeps discovering and ingesting cited work
- Argument graph: canonical claims with typed evidence across papers (establishes / corroborates / contradicts)

One substrate, many ways in

- Papers, patents, drafts, web pages, notes, email — everything lands as *chunks* (2.6M); each gets a summary (token-efficient reading, for humans too), keywords, an embedding, and a cluster
- Ask by *meaning*: semantic search over embeddings — including the HyDE trick: draft the answer you *wish* existed, embed it, find the passages that look like it
- Ask *literally*: grep-style lexical and verbatim modes over the indexed full text
- Ask by *category*: chunks are clustered for browsing and navigation; finer role / topic categorizers are layering in
- Ask by *link*: walk the typed graph — citations and evidence edges, down to the paragraph
- Heavily indexed for interactive speed: a **55 GB** database, **24 GB** of it indexes — the vector store alone is 33 GB. Approximate nearest-neighbour (ANN) indexing is what keeps it interactive: the closest of 2.6M embeddings without comparing against all of them

Finding the paper: the corpus hunts for itself

- The citation graph does most of the finding: every read paper's reference list *and* its citing papers (Semantic Scholar, both directions) become candidates
- Anyone can want a paper with one call — a DOI mints a stub and the fetcher chases it: the human, a quest agent mid-argument, or a 3 am dream
- Web and deep-research searches surface titles worth holding; title-only requests park in a backlog for triage instead of guessing a DOI
- Wanting is idempotent: re-requesting a held or already wanted paper is a no-op — no duplicate hunts
- Coverage watch: new corpus text probed against standing claims, so gaps announce themselves

Find them, get them, index them all.

Getting the paper: eleven ways, best version first

- An 11-leg cascade per paper: publisher APIs (Springer, Elsevier, Wiley) → open-access aggregators (Unpaywall, Crossref, OpenAlex, EuropePMC, CORE) → arXiv → Semantic Scholar → paid content API *last* (~1¢/file); the stubborn rest queue for *assisted download* — link ready, human clicks
- Version of record beats preprint; free beats paid — the order of the legs *is* the policy
- Markup first: publisher XML / JATS / arXiv $\LaTeX$ source beat the PDF as the text source; PDF-only papers go through ML layout analysis + OCR (Marker) — equations, tables and figures recovered as text, the PDF kept for printing only
- Retraction gate *before* download — retracted work never enters the corpus
- Retraction status is wired downstream too: retracted work is downranked in search and *blocked* as a citation anchor at export — a draft cannot ship citing a retracted paper
- Never gives up: retries back off to monthly — “a paper can become OA later”
- After fetch, the cascade: chunk → embed → summarize → extract keywords — background worker passes; a paper arrives *searchable*, no human in the loop

Checking the paper: references, retractions, injections

- Every held paper's bibliography is parsed: entries split, fields extracted (regex first, a small LLM for the stragglers), matched to a DOI — and to our held copy when we have one
- Inline markers resolve: the [12] next to a number in the text answers “which paper is that?” in one call
- Provenance preflight: hand it a reference list, it flags retractions, amendments and metadata mismatches (Crossref)
- “List this paper's references” for the agent: the table and the web view exist; the MCP verb is one view away
- Prompt-injection scan on everything fetched from outside (web, news, email): *suspect* → bannered as untrusted, *high* → body withheld. Deployed this week; the email model-scored tier is built but waits on mail polling. External text is delimited as data either way — the scan is the alarm, not the protection
- Next: the same scan across ingested PDFs corpus-wide

Token-efficient literature machinery

- Fisheye TOC: cluster-drill any paper; read paragraphs, not PDFs; pointer-sized handles (pa203165~14..27)
- Discovery layer: per-chunk keywords + summaries, precomputed once, reused every query
- Chunk-role categorizers (default-off cascade)
- Coverage audit: “did we cite *all* the relevant work?”

chunks (excerpt)

ref_id, ord	position in paper
text, token_count	the paragraph
section_path	where in the paper
page_first/last	PDF anchor
keywords	discovery layer
handle	terse pointer

links: evidence edges, per-paragraph

Fisheye in practice: drill, don't read

```
get(id='pa47100~17..18')    Cu-based SAAs for CO2 electroreduction · Angew. Chem. '26 · 144 chunks
~15  various cu-based saas · dopant-mediated mechanisms · c-c coupling
~16  co2rr mechanisms · ai agent · targeted computational strategies
~17  "The energy descriptor demonstrates a linear correlation with the
activation energy barriers of rate-determining step for C2+ formation"
~18  "...identifies the newly discovered M1/Cu(111) (M = Y, Lu) SAAs and
Ni1Zn1/Cu, Rh1Zn1/Cu DSAAAs as promising candidates with high C2+ selectivity"
~19  integrated workflow · three stages · universal design principle
path: view='toc' 144 chunks → 3 clusters → ~15..20 → read 2 verbatim
```

- Sharp center, compressed periphery: the asked-for paragraphs verbatim, neighbors as one-line summaries — context without the token bill. Under the hood it's a graph traversal over chunks and links — assembled by code, zero LLM calls
- In-text citations resolve to *held* papers — one hop from a number to its source, for the LLM and the human alike
- Patents drill the same way (`kind='patent'`, 101 held)
- Next: resolved cites as one-line *summaries*, not bare names — judge a reference without opening it

Taproot: claims with receipts

- A claim is a first-class object: one canonical sentence, citable by a short id — identical claims *converge* onto the same hub instead of duplicating
- Support is multi-paper and paragraph-granular: evidence edges run paper → claim, one per supporting *passage* — two passages in one paper are two edges
- Attaching evidence is a reading act: an LLM verifier accepts a passage only if it actually supports the claim; rejections are remembered, never re-billed
- Evidence must come from the *meat* of the paper — “X was doped with Y and we measured Z”, not “it is known that...”. The hearsay gate now strikes references / prior-work / background sections and quotes carrying citation markers from the admissible pool (mint gate, next slides)
- Found the hard way: a claim whose support traced back to a review paragraph — whose own sources did not support it. A review paragraph cites *elsewhere*; accepting one as evidence launders an unchecked claim into a verified one

Taproot: trust at read time

- How much to trust a claim is *computed from its evidence each time you read it* — never stored as a number that can go stale. Strongest to weakest: passage checked against the full paper, then abstract-only, then author-vouched, then unverified, then contradicted. Once verified against the full text it stays top-ranked: the paper *was* read — a later “trust me” can’t unread it
- Presented claim-first: the sentence up front, evidence on demand, and one jump lands *in the right section of the paper* (deep search) — no PDF, no scrolling. Cycle time is where the human loses focus, so the mechanical middle steps are eliminated
- The same trust function feeds the editor badge and the .docx / L^AT_EX exports — one truth, three surfaces

The argument graph

- Typed edges: establishes / corroborates / contradicts (paper $\rightarrow$ claim); refines / contradicts (claim $\leftrightarrow$ claim)
- Originators are derived, never guessed: citation edges *among the supporters* decide who was first; no evidence, no originator — just an honest coverage note
- Over-merge is the one forbidden direction: unconfirmed matches go to human review, and the live gate requires *zero* false merges
- Coverage watch — “did we cite all the relevant work?”: every newly embedded paragraph is probed against every claim; near-misses queue for evidence review
- Status: producers live — 1,552 canonical claim hubs, 2,059 evidence edges in prod, and the publication surface on top (next two slides)

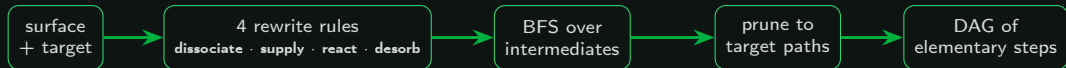
Nanopublications: claims signed to the world

- A claim hub that survives the gates becomes a *nanopublication*: one sentence + provenance + grounding (DOI, PDF hash, verbatim quote) as a tiny signed artifact — verifiable *offline* by a third party, no server of ours up
- The ladder ratchets frozen-ness: candidate → reviewed (string freezes) → signed (bytes freeze) → anchored (OpenTimestamps, Bitcoin calendar) → published (public forever) — every rung before publish is reversible
- Two keys, two meanings: the bot key is non-attesting; the human attesting key loads only through an interactive door and must match the signer's registered ORCID iD — “signed” means *a person checked*
- New contradicting evidence walks the ladder *down*: a reviewed or signed claim reopens to candidate; past the anchor, a human must supersede or retract
- Live since 2026-08-24: first claim published to the public nanopub registry (w3id.org trusty URI, OTS-anchored); 3 more signed, 138 candidates queued behind the human doors
- Mirror of everyone else: ~89k published nanopubs from the global registry cached + hash-verified locally, with a concurrence alert when someone else asserts one of our sentences

The claim gates: what a sentence survives to go public

- **Admission:** only papers, patents, EDGAR filings and datasheets ground evidence — a draft or memory never can; the passage must be asserting prose; an LLM verifier accepts it only if it supports the claim; a dialectic argument without an inline evidence handle never enters the record
- **Placement:** a sub-confidence duplicate goes to human review, never auto-merges (live eval: *zero* false merges); *contradicts* is written only at high confidence — it blocks someone's publication
- **Mint** — mechanical validators, no LLM in the path: verbatim quote in exactly one stored chunk · unique search snip · DOI + PDF hash · hearsay struck (reference / prior-work sections, quotes with citation markers, review-titled sources, papers we don't hold) · quantity bounds · fields contained in quotes · authoring model named · live *contradicts* blocks · drift (sentence re-hashed) · atoms sign before compounds
- **Type discipline:** a claim without a quote is blocked — a hypothesis *with* one is blocked too: it carries motivation + a discriminating experiment (≥ 2 independent source papers), never evidence
- **Publish preflight** — no mute button: every evidence edge machine-verified against the *live* sentence or human-signed-off interactively; trust allowlist — a bot signature alone publishes nothing; the POST is interactive + live-flagged + zero blocking issues
- Still judgment, not code: commensurability of a compound's binding; adjudicating a reopened dispute; retraction past the anchor

autocatpath I: auto-generating the reaction network



- Species are molecular *graphs* on the surface; four rewrite rules generate every elementary step: break a bond, stage an adatom ($+H^*/+O^*$), bond it, or let a closed-shell fragment leave
- Intermediates found by breadth-first search with canonical-graph dedup, valence and atom-budget caps — every step strictly makes progress, so the network is provably a DAG
- Geometries materialised per intermediate with a shared atom ordering across each step — exactly what NEB interpolation needs
- Curated templates remain available when the chemistry is known

autocatpath II: spend barriers only where they matter

- Relax every intermediate (BFGS, force + energy convergence) with ML potentials; barriers by climbing-image NEB on IDPP-interpolated bands, auto-retried on non-convergence; coadsorbed fragments by default
- Routes ranked by *energetic span* (Kozuch–Shaik); unrefined steps count as optimistic lower bounds, so ranking works before every barrier exists
- Best-first NEB scheduling: refine only the step that could change the winning route; stop when no rival route can win — barriers are bought, not sprayed
- **Dissolving tether**: each adsorbate atom on a spring to the slab, stiffness ramped to zero across relaxations — strays are pulled back into the well, yet every reported number is spring-free
- Uncertainty pooled across seeds *and* potentials (MACE / CHGNet / UMA / GRACE); CHE post-processing gives limiting + operating potential (V vs RHE)

autocatpath, decoded

ML potential	a neural net trained to reproduce quantum-chemistry (DFT) energies and forces — orders of magnitude cheaper, slightly wrong; we run <i>four</i> and let their disagreement be the error bar
MACE · CHGNet · UMA · GRACE (CI-)NEB	the four: independent architectures and training sets, so their errors don't correlate nudged elastic band: stretch a chain of geometries between two states and relax it; the <i>climbing image</i> slides to the saddle point — that saddle <i>is</i> the barrier
CHE	computational hydrogen electrode: every $\text{H}^+ + \text{e}^-$ step shifts by $-eU$ — potential dependence without simulating an electrode
energetic span	the one effective barrier of the whole cycle (Kozuch–Shaik) — what actually limits the rate, not the tallest single step
U_L (V vs RHE)	limiting potential: the least negative potential at which every step runs downhill; quoted against the reversible hydrogen electrode

autocatpath III: three audiences, three outputs

- **For the human:** a self-contained offline HTML report — clickable energy profile linked to rotatable 3-D structures, gallery, collapsible methods
- **For the workbench:** results land as an interactive pathway ref in précis — DAG, profiles, cross-model compare, dossiers, U-slider
- **For the quest loop:** a scalar summary (rate-limiting barrier, span, U_L , selectivity / trap / poison margins) harvested automatically as quest measures
- Every run: `methods.md` + a provenance snapshot — the write-up paragraph is generated, not reconstructed

Beyond one code: the simulation substrate

- Pathway explorer UI + dossiers inside précis (tier ladder, potential lever, lineage/dedup)
- Open catalysis DBs imported locally (OC20 vs AQCAt25 pick pending)
- Python simulation containers (sandbox-run; built, dark)
- CFD and CAD-for-CFD geometry via a cascade of simulators matched to each order of magnitude
- National HPC (ICHEC) as the big-compute rung: ML-potential sweeps and DFT where the local boxes stop

LLM hands on the model

- Whole simulations by run spec: edit the spec → run → profile / DAG / thumbnails come back
- A slab is a *typed design*: the LLM authors ops and reads probes, never pixels — pattern proven on the CAD and PCB kinds first
- Semantic diff: edit, then ask what changed — bonds formed / broken, atoms displaced — modify the model, *see what it did*
- Guardrails: impossible geometry hard-rejected before any GPU spend; cheap local relax first, cloud last resort
- Designs cite their reasons: the structure assembles its own literature query; provenance links say *why* it exists

navigate / inspect

```
get(view='neighborhood'|'bonds'|'find')
get(view='plane'|'rings'|'path'|'pov')
get(view='diff')    what changed
get(view='validate') is it physical
get(view='literature') why it exists
get(view='poscar'|'cif'|'extxyz')
```

write: edit(ops=[...])

```
slab  seed a whole fcc(111) surface
add_atom · set_element · vacancy
displace · add_bond · remove_bond
constrain · eye · measure
```

slab mirrors autocatpath's build_slab — identical atom order, so a design injects straight into a barrier run.

PCB maker: the same recipe on electronics

- A circuit is already a graph: the board lives as a typed netlist + placement IR the LLM reads as structure, never pixels — the CAD / slab pattern, applied to electronics
- **Netlist checking** is where the LLM shines: it walks the logical signal trace pin by pin and questions intent; the eyes (ratsnest, crossings, proximity) plus geometric and per-net connectivity DRC give it exact, unit-tested answers — no LLM in that loop
- **Datasheet reading**, the other strength: datasheets ingest through the full paper pipeline (fisheye TOC, in-document search), link *datasheet-of* to their parts — “check pin 7 against the datasheet” is one call, and a datasheet can ground evidence like a paper
- Joint place+route: simulated-annealing placer + occupancy-grid maze router down to zero DRC violations — IPC-2221 track widths, plane pours, via shoving, silkscreen; JLCPCB-native parts catalog
- Export only at the edge: gerber fab bundle + BOM/CPL from one view — the design never leaves the relational graph until fab
- Toward the first ordered board: reference boards route clean; the topological place+route write-up is a living précis draft

Mechanical design: one space plan, atoms to buildings

- One scale-agnostic design kind: float64 metres throughout, so a photoswitch and a footbridge are the same object — the one unit crossing in the system is the Å multiply where an atomic block binds real atoms
- A design is deliberately *suggestive* and hardens monotonically — “a fork about this size, on a hub that goes through a wheel so the wheel can turn”. Every field past a name is optional; absence is *reported*, never rejected
- Six levels, each valid alone: drop everything above level k and what is left still checks
- Invariants are *declared*, never reverse-engineered from geometry — joints as kinematic class $\times$ mechanism, tolerances as relations between named measures on a pinned datum, loads as objectives
- Arrays and instances stay a tree; realization never flattens it

L0	block graph, ports, intent
L1	envelopes + pose (metres)
L2	joints, tolerances, loads
L3	realized solids <i>or</i> atoms
L4	fit, stack-up, DRC
L5	fabrication plan, export

read

```
view='tree'|'block'|'ports'  
view='validate'|'drc'|'clearance'  
view='stability'|'sweep'|'fret'  
view='bom'|'fasten'|'print'|'fab'
```

Discrete block states carry a photoswitch and a macro bistable on one mechanism; `view='sweep'` asks whether anything collides in *any* of them.

What the structure has to answer for

- A verdict, not a picture: one SVD of the equilibrium matrix gives Maxwell–Calladine $m - s$, self-stress sign-checked against each member's tension/compression capacity, then the Pellegrino–Calladine second-order test — *rigid / mechanism / prestress-stabilized*
- Dogfood: a printed unicycle, 12 tension-only spokes at 600 N verified as a self-stress state — and the checker still called the wheel a first-order mechanism, correctly: radial lacing transmits no torque
- Generators kept apart from checkers: force-density form finding solves the equilibrium geometry and writes poses back stamped *proposed*; a human-set pose is contract and moves only under explicit authorization
- Down to the shop: ISO 273 clearance and tapped holes with the grip stack-up walked along the screw axis; a printed boss refuses a cut thread and names the four strategies; a swept driver envelope says *which* tool reaches; printed solid + build-frame search, STL/3MF only at the edge
- The library answers back: `wants={...}` ranks every block against a wishlist (Pareto tie-break, never a filter — each row shows its misses); `compose={...}` enumerates switch + spacer stacks into a stroke/span box
- Engines ahead of their callers: 3D SIMP topology optimization (overhang filter, gyroid fill), active-set tension-only/contact equilibrium, and nonlinear continuation that reports a *real* snap-through barrier — all tested, all screening-tier by construction, none yet wired to a view

Curved carbon by notation, not atom lists

- A .hx file is a string you can paste into a paper: sp^2 carbon as *defect placement on a hexagonal lattice* — sheets, (n, m) tubes, cones, C_{60} , holes, fused seams, nanobuds — atoms derived deterministically
- The topology carries its own proof:
 $\sum_n (6 - n)P_n + B = 6\chi$ per sheet component, so a shape that cannot close is an *error at compile time*, not a relaxation that quietly fails
- Checks before atoms — valence, ring sizes, port symmetry, overlapping cuts — and fidelity='check' reports without minting: the agent's edit/check loop
- Wired in as an se atomic generator: the spec text is the regeneration input, the coordinates a declared preview — physics is the tier above
- Next: *negative* curvature — junctions, tori, periodic P/D/G cells, schwarzites as tubes on a skeletal net — the counting law already carries them

```
# a post fused into a sheet
lattice: element=C sigma=1.42
substrate: sheet(20,20)
  - hex(0)@(9,9,A):0
post: tube(6,0,len=4)
substrate.hole -fuse k=0-> post.in
registry: post.in == post.out
terminate: substrate.rim = H
```

Twelve worked examples ship with the notation.

From corpus to manuscript

- Chunk-native drafts: every paragraph keeps its evidence links as it is written and rewritten
- Citations generated from taproot: one call per source → BibTeX / RIS / EndNote
- .docx export embeds an EndNote *traveling library* (reverse-engineered Cite-While-You-Write fields) — co-authors open Word and the references are simply there
- Figures extracted at ingest with page anchors — image sources are cited like text
- Third-party figures tracked by rights: origin (original / own graph / third-party) + permission with expiry — an uncleared figure *fails the export*, like a bare cite fails review
- A draft citing a signed claim gets a “Published claim artifacts” appendix — frozen sentence + trusty URI — in both exporters; unminted claims leave the export byte-identical
- Living outputs: this deck and the paper, regenerated as the corpus moves

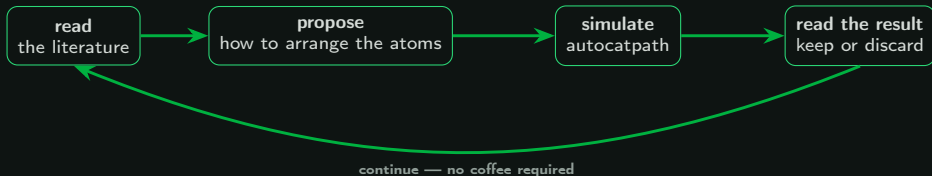
Grounding the human I: the other context window

- Human working memory is the scarce resource: recall is slow, screens are small — optimize for it like we optimize tokens
- The fovea — the only sharp patch of the retina — covers a word or two at reading distance, far *smaller than the screen*; “seeing the page” is a behavioral trick of constant eye jumps. Design for the sharp spot, not for the page
- Taproot presentation: claim first, evidence on demand, jump straight to the section
- Terse pointer handles keep human–machine exchanges short on both sides
- Turn-taking: the system paces itself instead of flooding the channel

Grounding the human II: sensing and shaping

- Experts recall vast latent detail in a blink; elsewhere the same person's understanding is sparse — and telling which is which is genuinely hard, from outside *and* inside
- Anki integration: the system's running estimate of what the human actually retains — sensing the human is hard, and that is exactly why it matters
- Morning / evening briefs: podcast *and* text — listen on the commute, grep it back out later
- Weekly / monthly digests of newly ingested work
- The perpetually managed paper: semi-annual updates, always current — this deck is its seed

Untiring: the researcher loop, run all night



- What a researcher does — but it doesn't stop at 6pm: quests keep the loop turning overnight
- Honestly: *autonomous exploration* is the flaky part — it still fails a lot. Everything around it is surprisingly stable, and failures are usually one Claude session away from fixed (*the Shannon does flow past UL*)
- Work *parks itself* on a missing paper or pending feedback and resumes alone when it arrives
- Review tiers (nursery → structural → deep) gate what sticks; anti-spin guards + budget breaker bound the night
- Dreaming pass: far-apart points in embedding space, connected through a scientific lens — inspiration, never citation

Dreaming: speculative connections on idle compute

```
DREAM CYCLE -- no user is reading. Fact-anchored speculative connections, not introspection.
```

```
1 sample your own old memories -- diverse, not recent (view='dreamable')
```

```
2 sample papers the same way; pick one you don't recognise and read it
```

```
3 pull one anchor from another kind: patent · oracle · web · deep research
```

```
4 write  $\geq 2$  non-obvious cross-kind connections -- each must pass through a fact
```

```
5 request papers you wish you had (stubs are chased and auto-fetched later)
```

```
6 verify your own work: every cited handle must resolve; promote survivors, keep failed hunches as hunches
```

```
me205091 · tier:synthetic-insight (survived its own review)
```

"NrfA wins its six-electron nitrite $\rightarrow$ NH₃ selectivity by *compartmentalising delivery*, not by tuning an exposed site: the substrate binds in a secluded pocket on a lysine-ligated heme, with four further hemes wired in as an electron relay [pa202554, pa3039]. That is the solved, biological form of the critique my own earlier memories aimed at the quest's Pd-slab Pareto — the rate limit is local proton and electron *supply* to a buried site, not the intrinsic barrier of a bare slab. It argues for the confined-water / proton-shuttle overlayer as the synthetic analogue."

- Runs on idle local compute, ~every 15 minutes — 9,222 dream memories so far
- One anchor seeded from an active quest, the other free-roads — cross-domain by construction (paper $\leftrightarrow$ patent $\leftrightarrow$ web)
- Every dream adds edges: the link graph densifies, so tomorrow's search finds paths today's couldn't

Division of labor: what the LLM does, what code does

- **The LLM as scientist** — judgment work: reads papers, forms hypotheses, decides how to arrange the atoms, weighs what the simulation returned
- **The LLM as coder** — builds the system itself, at build time: gated by tests, one worktree per session (the dev-loop slides)
- **Code serving the LLM** — at runtime, deterministic machinery does the heavy lifting: fisheye context assembly (a graph traversal), search + ranking, autocatpath's path search and barrier pipeline, the worker cascade — cheap, reproducible, zero tokens
- The rule of thumb: what must be *right every time* becomes code; what needs *judgment* stays with the LLM
- The boundary migrates: a transformation the LLM finds itself doing twice gets written down as code the third time

A deliberately small tool surface

- Seven verbs, one `kind=` discriminator — the entire MCP surface an agent ever sees
- Skills are *embedded*: agents discover the manual semantically (`search(kind='skill', q=goal)`) instead of having it stuffed into every prompt
- Small surface = small schemas = tokens saved on *every single* tool call

the whole API

search	find — lexical + semantic
get	read a ref, or compute
put	create / import / annotate
edit	anchored region edits
delete	remove (recoverable)
tag	classify along typed axes
link	typed relations between refs

× `kind=` paper, patent, draft, quest, cad, pcb, protein, email, ... (53)

One ACID database under everything

- Everything is a ref in PostgreSQL: papers, claims, quests, jobs, gripes — **78** tables, one transactional truth
- ACID is what makes *many agents* safe: atomic ingest, advisory-lock claims, lease tables — relations never clobbered, work never done twice
- Append-only where it matters: body chunks are never mutated; migrations are forward-only
- Vectors live beside the rows (pgvector) — no second database to drift out of sync
- Multi-user by construction: role-separated access through a pooler — scales to a small team without redesign

key tables

refs	every object
chunks	2.6M paragraphs
chunk_embeddings	the vectors
links	typed relations
ref_tags	status / axes
jobs	the factory ledger
ref_events	full audit trail

One router, every model

- Capability-tiered routing: each call states what it needs; the router picks the cheapest model that clears the bar
- Chains: local-first with cloud fallback per tier; an 11m catalog kind tracks variant-precise offerings
- Proprietary content pinned to local models — sovereignty enforced as a routing rule, not a promise
- Status: built end-to-end — awaits its first full deploy and a debugging pass
- National-HPC burst: biggest local models via Slurm on ICHEC (built, dark — awaiting cluster access)
- Provider cascade: local → OpenRouter → Anthropic with automatic failover

Runs on our own silicon

- A small office fleet — a few Macs and a pair of DGX Sparks sharing one NAS — is the factory floor
- Large open-weight models served in-house carry the bulk of the load; cloud only where capability demands it
- Economics instrumentation: tokens per answered question, \$/day local vs cloud (probe in flight)

Grounded *and* cheap enough to run all night — that is the pitch.

(There is more compute at home than in the entire lab.)

Evaluation: how we will know it works (proposed)

- Today: every artifact still gets a manual end-to-end review before it leaves the building — *no slop ships*; the metrics below complement that, they don't replace it
- Citation integrity: % of claims whose anchor resolves, + spot-audit “does the paragraph actually support the claim”
- Coverage: recall against must-cite sets built from review articles' bibliographies
- Simulation: autocatpath vs published barriers / limiting potentials on known systems; cross-potential agreement already a built-in error bar
- Cost: tokens per *grounded* claim, vs plain RAG
- Human loop: Anki retention as the measurable of the human-side pipeline

Dogfooding: the system reports on itself

- This deck's numbers were pulled live from the production database; the dots reflect actual deploy state
- The system files gripes on itself — **391** filed, **275** in the last 30 days — 242 of those already closed, **29** still open; agent-transcript mining turns LLM confusion into documentation fixes
- The paper will be a living précis draft — the artifact demonstrates the method

Two manuals I: skills for the research agents

- The product's manual lives *in* the product: **143** skills served as refs — the agent literally asks `search(kind='skill', q='how do I ...')` at the moment of need
- Token economy: nothing preloaded — a session carries only the recipes it actually pulled, not a 143-chapter manual in every prompt
- They teach research conduct: how to cite, how to attach evidence, per-kind recipes, whole tool-call sequences (toolpaths)
- Self-improving: transcript mining turns agent confusion into skill fixes; gripes point at the gaps
- The agents read the same manual they help improve

Two manuals II: skills for the coder

- The repo carries a *separate* manual for the agents that build précis: ship workflow (/land, /go), test discipline, backlog care, conventions-that-bite
- Kept deliberately apart: “how to cite” and “how to land a branch” confound easily — different readers, different manuals, never mixed
- One source of truth per fact: codebase map → package docstrings → glossary; generated indexes, never hand-edited
- 19 sized agent roles (cheap gophers → deep reviewers) — delegation is the cost lever
- Memory that survives context compaction: threads, runbooks, gotchas — the coder wakes up knowing what bit it last time

From N=1 to a lab

- Already multi-agent with one shared, provenance-carrying corpus — adding humans scales the readers, not the truth
- Per-member human loop: own Anki state, own briefs, own reading queue over the shared substrate
- Specialization by quest: members own quests, agents specialize per quest — **shared knowledge stays in the argument graph and term registry**, so expertise diverges without the ontology forking
- Onboarding: taproot + the brief backlog — catch up in weeks, not a year of reading

Inside Claude Code: the harness that writes it

- **Given** to the coder: **19** sized agent roles (8 mechanical, 11 for bounded changes; architecture stays with the main loop), 6 one-word workflows (`/land`, `/go`, `/next...`), **61** scripts
- **Enforced, not requested:** **20** hooks fire *inside* the agent's tool calls at 5 lifecycle points — blocking writes to production, edits to sealed migrations, commits on `main`, secret reads, blind `git stash`
- The softer ones redirect instead of blocking: a bare-symbol `grep` becomes a structural query; a stale map or an unexplained bug triggers the triage path
- **Grown** by the coder: **93** memory files — threads, runbooks, gotchas — reloaded every session, so it wakes up knowing what bit it last time
- **Written down:** 12 conventions, 16 runbooks, **198** open backlog items, and one router document that points at the rest
- Same shape as *précis* itself: a manual and tools handed over, memory and links accumulated — the symmetry is deliberate

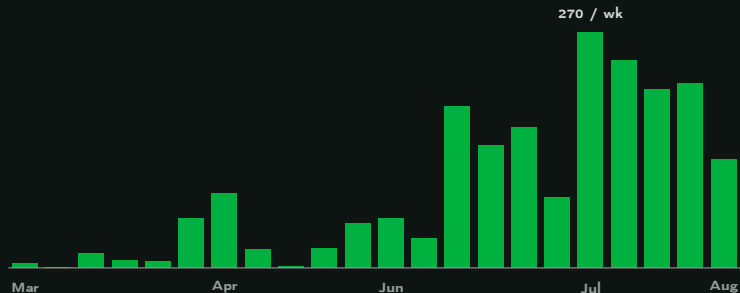
The dev loop behind the curve

- Shipping is one word: `/land` = `commit` → `sync` → containerized gate (ruff, mypy, tests) → squash to main; `/go` adds the full suite + cluster deploy
- Branching: trunk-based — one short-lived branch per session worktree, squashed onto an always-green `main` (one commit per ship, linear history); no `develop` or release branches, a branch dies when it lands
- Many Claude sessions in parallel worktrees; merges are lock-free compare-and-swap pushes; finished worktrees reap themselves
- Inner loop runs only the tests the change touches (testmon); the full suite gates every deploy
- Backlog: one file per item (**198** open), deleted on ship — git log is the only done-log; gripes catch the residuals
- The dev context is budgeted like tokens: CLI output compressed 60–90%, semantic code index, memory that survives compaction

Coding with an LLM: where the human time goes

- Almost no time reviewing code line-by-line — attention goes to *data structures*: tables, schemas, invariants, interfaces. Get those right and the generated code follows
- Hardly any object-oriented design to argue about either — the codebase converged on plain data + functions, a shape the model handles very well
- Stability comes from the harness, not from trust: lint, typecheck and unit tests gate every merge; red never ships
- The recurring failure mode is small: occasional web-UI JavaScript glitches — paste the browser console into the session, fixed in 1–2 turns
- Net: very stable. Five months of daily LLM-written commits and the system stays up

Five months of effort



- **1,989** commits in 22 weeks — précis repo alone; autocatpath keeps its own history
- Eras read off the graph: MCP core → drafts + web UI → quest factory, LLM router, taproot → deploy & hardening
- Sustained by the dark factory: overnight workers + review gates, not heroics

Status scoreboard

- Ingest + search + taproot
- Nanopublications: gated mint → sign → OTS → registry (first one public)
- Fisheye TOC / discovery layer
- autocatpath core (NEB, CHE, uncertainty)
- Pathway explorer + dossiers
- Quest / todo factory + review tiers
- Local LLM fleet (Macs + Sparks)
- Morning / evening briefs
- Citation export incl. EndNote traveling library
- Slab structure kind (ops, probes, diff)
- se: one design kind atoms-to-buildings (stability, fasteners, print, library search)
- hexfold: curved sp^2 carbon from notation
- One router, every model (first deploy + debug pending)
- Argument graph · categorizers · Anki
- Catalysis-DB import · sim containers
- Economics instrumentation
- Topology-optimization engines (SIMP, snap-through) — tested, not yet wired to a view
- Negative-curvature carbon: junctions, tori, periodic cells, schwarzites
- ICHEC national HPC: LLM burst (built, dark); sim / ML-potentials / DFT to follow
- Coverage audit · evaluation suite
- Digests · living paper
- Multi-member lab mode

Next

- Now: economics numbers (probe out), first evaluation metrics wired into the scoreboard
- Decisions pending: OC20 vs AQCat25 import; journal target (recommendation: *Digital Discovery*)
- Paper: living draft seeded from this deck; autocatpath JOSS companion
- Asks: [hardware / time / student — fill in]

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